

Customer Churn Prediction Project: From Data to Insights

Welcome to an exploration of our end-to-end customer churn prediction project. This presentation outlines the three critical phases that transform raw data into actionable insights, designed to boost customer retention and drive sustainable growth.



Three Phases of Our Project



Phase 1: Data Preparation

Extract, Load & Clean Data for Analysis



Phase 2: Orchestration

Automated Pipeline Management & Scheduling



Phase 3: Visualization

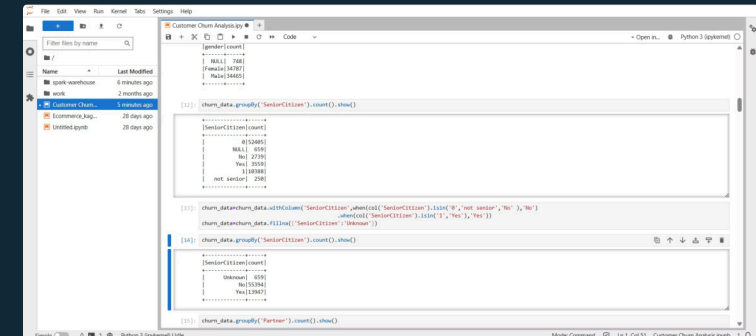
Actionable Insights & Interactive Dashboards with Power BI

Phase 1: ETL for Data Cleaning – Laying the Foundation

Raw customer data often contains noise, missing values, and irrelevant features, which can significantly hinder model performance. Our rigorous ETL (Extract, Transform, Load) process is designed to clean, transform, and prepare this data, ensuring it is pristine and ready for sophisticated modeling.

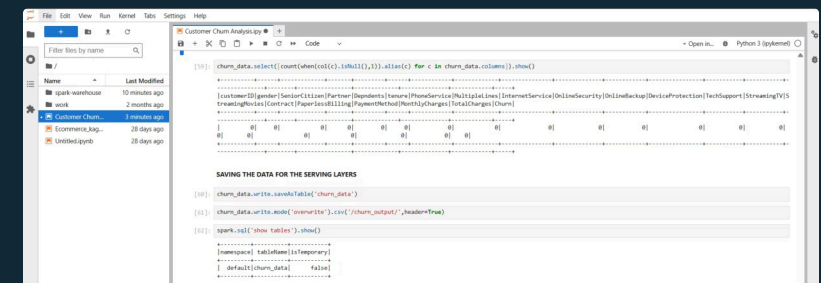
- Collected customer data from Kaggle and loaded it into HDFS using my custom cluster simulation environment
- Applied data cleaning and transformation using Apache Spark
- Handling missing values in an efficient way.
- Stored the processed data in MSSQL Server.

The result is a high-quality dataset that dramatically boosts model accuracy and reliability, forming a solid foundation for all subsequent phases.



```
from pyspark.sql.functions import count, when, isnull, col, concat_ws, concat

# [12] churn_data_groupby('SeniorCitizen').count().show()
# [13] churn_data_groupby('SeniorCitizen').count().show()
# [14] churn_data_groupby('SeniorCitizen').count().show()
# [15] churn_data_groupby('Partner').count().show()
```



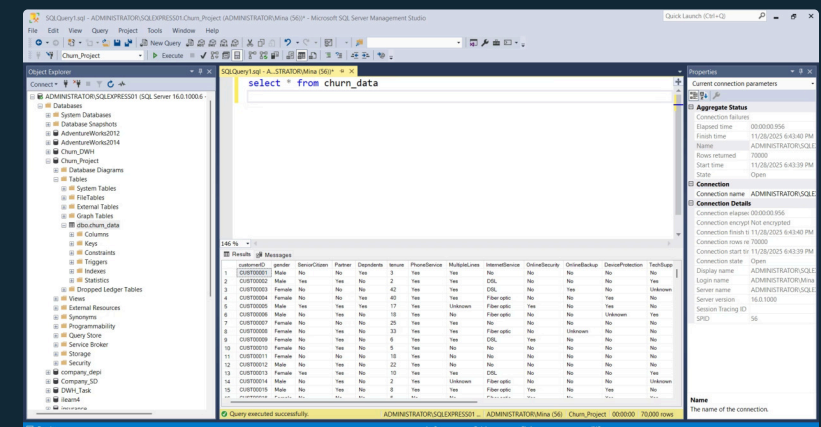
```
churn_data.select(count(when(col(c).isnull(),1)).alias(c) for c in churn_data.columns).show()

SAVING THE DATA FOR THE SERVING LAYERS

churn_data.write.mode('overwrite').csv('churn_output',header=True)

spark.sql('show tables').show()

churn_data.write.mode('overwrite').parquet('churn_output',header=True)
```



customerID	gender	SeniorCitizen	Partner	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnbillPayment	DeviceProtection	TechSupport	StreamingServices
1	Male	No	No	1	Yes	No	Yes	No	No	No	No	No
2	Female	No	No	2	Yes	No	Yes	No	No	No	No	No
3	Female	No	No	42	Yes	No	DSL	No	Yes	No	Unknown	No
4	Female	No	No	43	Yes	No	Yes	Yes	Yes	Yes	Yes	No
5	Male	Yes	No	17	Unknown	No	Yes	Yes	Yes	Yes	Yes	No
6	Female	No	No	16	Yes	No	Yes	Yes	Yes	Yes	Yes	No
7	Female	No	No	25	Yes	No	Yes	No	No	No	No	No
8	Female	No	Yes	33	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
9	Female	No	Yes	5	Yes	No	Yes	No	No	No	No	No
10	Female	No	No	18	Yes	No	DSL	No	No	No	No	No
11	Female	No	No	10	Yes	No	Yes	No	No	No	No	No
12	Male	No	Yes	22	Yes	No	No	No	No	No	No	No
13	Female	No	No	10	Yes	No	DSL	No	No	No	No	No
14	Male	No	Yes	2	Yes	No	Yes	Yes	Yes	Yes	Yes	No
15	Female	No	No	5	Yes	No	Yes	Yes	Yes	Yes	Yes	No

The screenshot displays the DagsHub interface for a DAG named 'XGboost_model'. The top navigation bar includes a 'Dag Run' section with a 'chum_id' and a 'Task' section with 'XGboost_model'. The main area shows a DAG graph with four tasks: 'get_data', 'data_preprocessing', 'XGboost_model', and 'starSchema_load'. All tasks are marked as 'SUCCESS'. The 'XGboost_model' task is highlighted with a red box. The right sidebar provides details for the DAG run, including the operator '@task', start and end dates, and a log snippet showing the model's accuracy and a warning about unused parameters.

DAG Run Details:

- DAG Name:** XGboost_model
- Operator:** @task
- Start Date:** 2025-10-29 02:11:00
- End Date:** 2025-10-29 02:11:03
- DAG Version:** v1

Log Snippet:

```

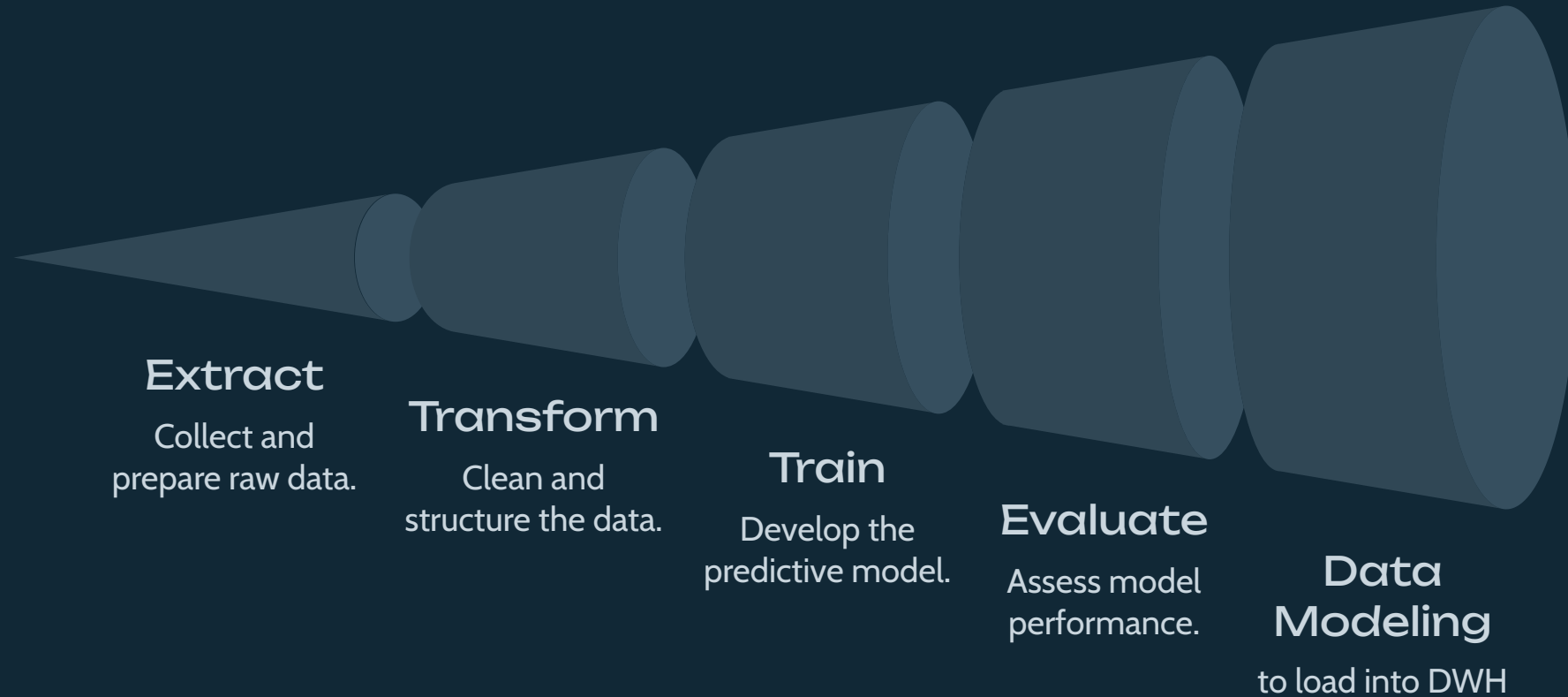
INFO - Model Accuracy on Test Set: 0.7845 source
INFO - Done. Returned value was: customer1

```

- Orchestrated the workflow using Apache Airflow DAGs to automate data extraction from MSSQL Server via Airflow Hooks

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- The screenshot displays the Microsoft Power BI Desktop interface. The top ribbon shows the 'Home' tab with various data source and transformation options. The main workspace shows a data model diagram with four tables: 'service_dim', 'customer_dim', 'fact_table', and 'contract_dim'. The 'service_dim' table is connected to 'fact_table' via a relationship. The 'customer_dim' table is also connected to 'fact_table'. The 'contract_dim' table is connected to 'fact_table'. The 'Data' pane on the right shows the list of tables and their columns. The 'Properties' pane on the right shows the 'Cards' view and options for showing database headers and related fields.

Airflow Pipeline Architecture

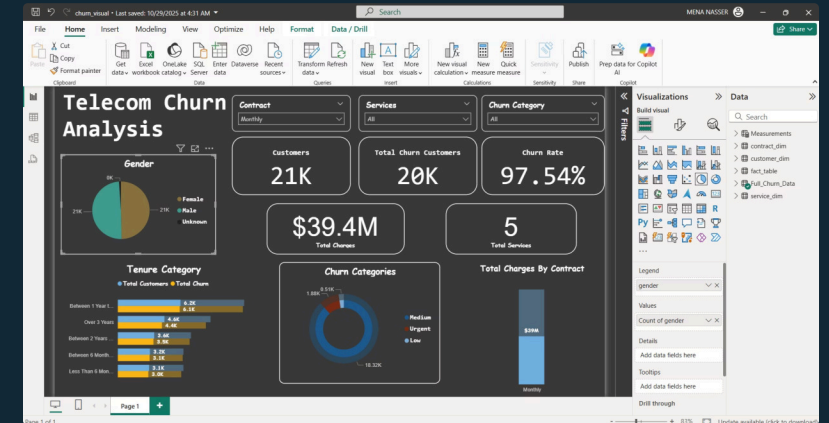


Automation ensures scalability and repeatability. This DAG (Directed Acyclic Graph) visually represents the streamlined flow of our AI model pipeline, highlighting how each critical step is connected and executed.

Phase 3: Insights Using Power BI – Turning Predictions into Action

Power BI dashboards are instrumental in visualizing churn risk segments, identifying key drivers of churn, and tracking customer behavior trends. These interactive tools empower business teams to make informed decisions.

- Business users can easily identify **at-risk customers** and tailor retention strategies proactively.
- Built an interactive Power BI dashboard to visualize key insights and support a retention strategy based on churn probability.
- Interactive reports and customizable filters allow for deep dives into specific customer segments, empowering data-driven decisions and proactive customer engagement efforts.



The screenshot shows a Power BI data table titled "Full_Churn_Data". The table displays customer data with columns for PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, ChurnProbability, Churn, ChurnCategory, RetentionStrategy, TenureCategory, and services. A red box highlights a specific row of data.

PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	ChurnProbability	Churn	ChurnCategory	RetentionStrategy	TenureCategory	services
Yes	Electronic check	44.12	\$882.0004	0.74950471183014	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	38.89	\$77.78	0.71946763923096	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	28.87	\$1,794.4489	0.61326026203507	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	48.19	\$1,252.94	0.74926570320307	Medium	10% Discount	Between 2 Years to 3 Years	Between 2 Years to 3 Years	PhoneService
Yes	Electronic check	45.93	\$2,066.85	0.70476527775374	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	52	\$848	0.69754328380779	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	34.1	\$1,602.7	0.76128619903805	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	101.71	\$4,373.53	0.68122788326286	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	44.23	\$2,034.58	0.70702272855707	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	26.07	\$156.42	0.71937172628306	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	19.86	\$895.1004	0.78906832008122	Medium	10% Discount	Between 2 Years to 3 Years	Between 2 Years to 3 Years	PhoneService
Yes	Electronic check	44.77	\$2,154.75	0.70030100926252	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	37.39	\$1,458.21	0.65476499998105	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	29.34	\$762.18	0.701631096343994	Medium	10% Discount	Between 2 Years to 3 Years	Between 2 Years to 3 Years	PhoneService
Yes	Electronic check	18.47	\$265.52	0.75259170948919	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	36.98	\$998.4596	0.67451789636119	Medium	10% Discount	Between 2 Years to 3 Years	Between 2 Years to 3 Years	PhoneService
Yes	Electronic check	42.75	\$138.25	0.67906352043152	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	113.87	\$2,179.71	0.58209447658559	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	45.04	\$990.88	0.66071585512247	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	20.44	\$758.28	0.66804570083618	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	32.27	\$1,335.0701	0.711801160840716	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	18.25	\$201.85	0.54608882177592	Medium	10% Discount	Between 6 Months to 1 Year	Between 6 Months to 1 Year	PhoneService
Yes	Electronic check	32.7	\$1,568.6001	0.672062732167816	Medium	10% Discount	Over 3 Years	Over 3 Years	PhoneService
Yes	Electronic check	26.51	\$356.71	0.68101645159912	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService
Yes	Electronic check	48.44	\$1,532.64	0.68385862767277	Medium	10% Discount	Between 2 Years to 3 Years	Between 2 Years to 3 Years	PhoneService
Yes	Electronic check	23.53	\$188.24	0.688911278247833	Medium	10% Discount	Between 6 Months to 1 Year	Between 6 Months to 1 Year	PhoneService
Yes	Electronic check	21.19	\$217.85	0.68211073804477	Medium	10% Discount	Between 1 Year to 2 Years	Between 1 Year to 2 Years	PhoneService

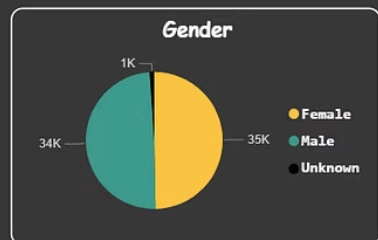
Telecom Churn Analysis

Contract
All

Services
All

Churn Category
All

Filters



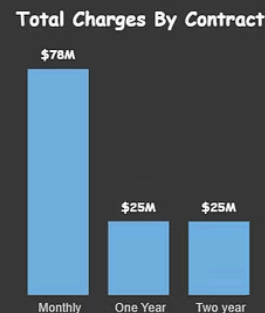
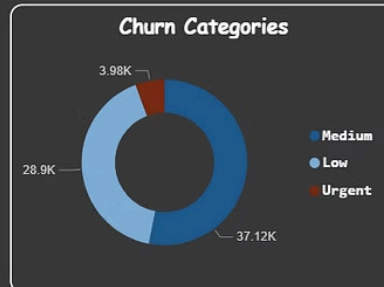
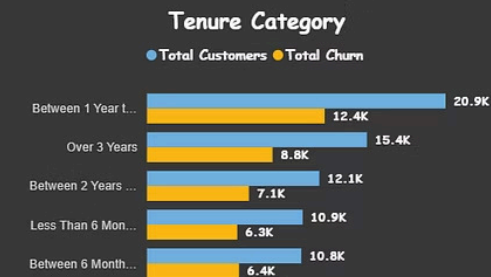
Customers
70K

Total Churn Customers
41K

Churn Rate
58.71%

\$128.4M
Total Charges

9
Total Services



Sample Power BI Dashboard

From data to decisions in real time. This snapshot showcases the intuitive design of our Power BI dashboards, making complex data accessible and actionable for all stakeholders.

Thank You

We appreciate your time and attention today. We are confident that this churn prediction project will empower your business with the insights needed to foster stronger customer relationships and achieve sustainable growth. We look forward to partnering with you on this journey!

